Lab 2 - AI^2 Product Specification

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1 Introduction

1.1 Purpose

This purpose of this document is to describe the requirements, design, and implementation of the AI^2 prototype. It is intended for developers to use as a reference while developing the prototype.

1.2 Scope

AI^2 is a web-based platform that seeks to alleviate these problems by managing high volumes of invoices quickly and accurately. It is designed to streamline repetitive invoice processing and allow AP team members to instead focus on high-risk invoices that demand greater scrutiny. This will benefit companies that must sort through a large number of invoices, as well as the vendors who will have their transactions processed much more efficiently.

The most critical features of AI^2 are the AI Confidence Scoring Module and the Queue Assignment Dashboard. Confidence scoring will allow the system to pick out invoices and purchase orders that may be mismatched, flagging the match for the AP team to review. The queue assignment dashboard will display invoice matches assigned to the current user, based on criteria configured by the administrators. The goal is to create a balanced workload among the AP team and ensure that no invoice is overlooked.

AI^2 must log all procedures for both internal and governmental compliance purposes. AI^2 will not process emails that are not invoice related, nor will it replace human reviewers.

1.3 Definitions, Acronyms, and Abbreviations

AI² (Artificially Intelligent Invoices): The proposed automated invoice processing system designed to streamline Accounts Payable workflows through AI-driven matching, classification, and routing.

Alteryx: A data automation and analytics tool that enables secure API connections and data workflows between applications.

AP (Accounts Payable): The amount a business owes for goods and services purchased on credit; typically due at intervals of 30, 45, 60, or 90 days depending on repayment terms.

APEX: A strongly typed, object-oriented programming language that Salesforce developers use to execute flow and transaction control statements on the Salesforce platform.

Atlantic Diving Supply (ADS): An American federal contractor company that provides equipment and logistics solutions to the Department of Defense, federal agencies, and first responders.

AWS GovCloud: A secure U.S. government-compliant cloud environment that supports sensitive data and regulated workloads in accordance with FedRAMP and DoD requirements.

CMMC (Cybersecurity Maturity Model Certification): A framework developed by the U.S. Department of Defense to assess and enhance the cybersecurity posture of contractors handling federal information.

ERP (Enterprise Resource Planning System): An integrated software system used to manage core business processes, such as finance, procurement, and supply chain operations.

FedRAMP (Federal Risk and Authorization Management Program): A U.S. government program that standardizes security assessment, authorization, and continuous monitoring for cloud services.

Graph API: A Microsoft API interface used to access and automate data operations within Outlook and other Microsoft 365 services.

Invoice: A document listing goods or services provided, including a statement of the amount due; a bill.

Oracle: The system of record used by ADS for enterprise data and financial management.

PO (Purchase Order): The official order list ADS sends to suppliers detailing the items or services requested.

Salesforce: A robust Customer Relationship Management (CRM) platform that allows businesses to manage customer relationships efficiently.

Supplier: A person or organization that provides goods or services needed by another organization.

Vendor: A person or company offering goods or services for sale.

1.4 References

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1.5 Overview

Section 2 of this document will describe the detailed components of AI^2, system interactions, real-world product and prototype functions, user characteristics, and constraints.

2 Overall Description

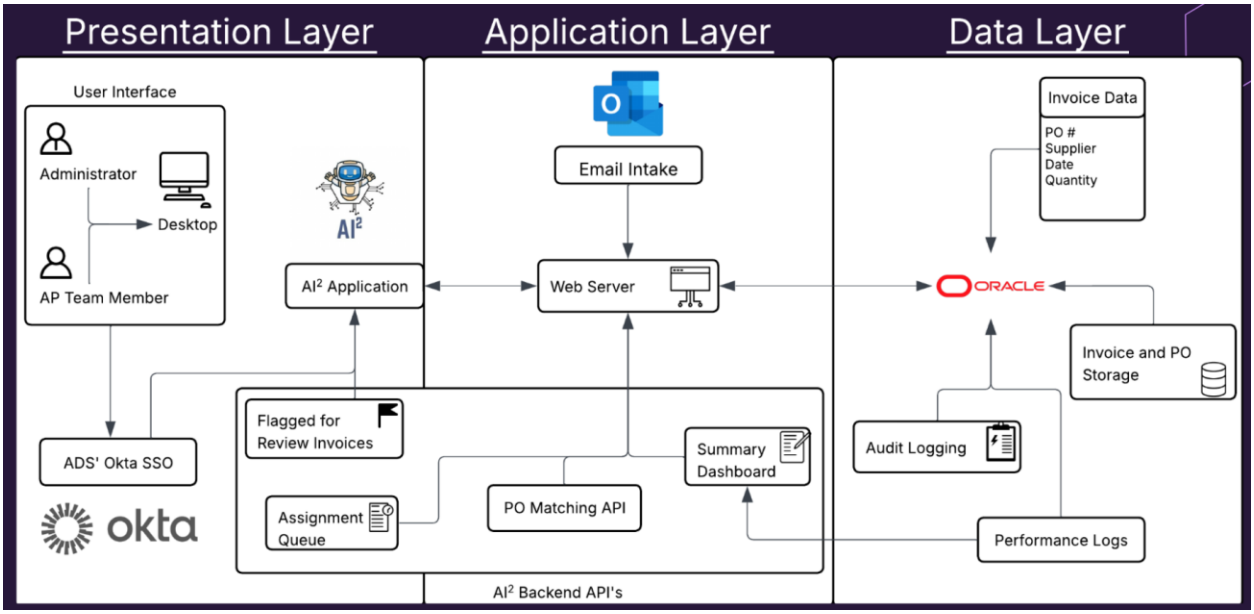
2.1 Product Perspective

AI^2 is designed for desktop computer use only. All users are authenticated through Okta SSO to ensure secure access and compliance. AI^2 connects to an Outlook inbox using Microsoft Graph API to pull incoming invoices, and to Oracle NetSuite through the Alteryx API to process the necessary data. From the desktop, authenticated users will view the presentation layer. The main components are the assignment queue and the low-confidence invoices that have been flagged for review.

The application layer handles the invoice extraction from the e-mails and transferring the results to the data layer and to the AI^2 backend. The backend is where the matching API will attempt to match an invoice to a purchase order and give a confidence score. AI model trainers will view the summary dashboard, which shows an assessment of the model's performance in extracting the necessary invoice fields and finding a PO match. The data layer stores all extracted invoice and PO data, as well as an immutable audit log indicating changes to the AI model and the database.

Figure 1

Major Functional Components Diagram



2.2 Product Functions

The prototype will focus primarily on features that pertain directly to invoice processing. These include the data extraction, marching algorithm, confidence scoring, invoice quality analyzer, and AI model configuration. The prototype should demonstrate the ability to automatically and accurately process invoices, and then present this information to AP team members.

Table 1

Prototype Feature Table

Feature	Prototype
Outlook Email Integration (Graph API)	Eliminated
Invoice Parser and Data Extractor	Fully Functional
Automated PO Matching Algorithm	Fully Functional
Invoice Quality Analyzer	Fully Functional
Queue Assigning and Routing	Partially Functional
Queue Assignment Dashboard	Partially Functional
Model Training and Deployment Pipeline	Fully Functional
Model Version Control	Fully Functional

AI Performance Dashboard	Fully Functional
Invoice Data Anonymization Pipeline	Eliminated
Admin Configuration Dashboard	Partially Functional
System Health and Performance Dashboard	Partially Functional
Error Logging and Recovery Module	Fully Functional
Admin System Alert Notifications	Partially Functional
Automated Supplier Notifications	Eliminated
Automated AP Team Reminders	Partially Functional
AI Confidence Scoring Module	Fully Functional
Continuous Learning from Human Interaction	Fully Functional
Human Review Panel	Fully Functional
Encryption Verification Tool	Eliminated
AWS GovCloud Integration Tool	Eliminated
Audit Log Management System	Partially Functional
Compliance Changelog System	Partially Functional
Auto Generated Compliance Report	Eliminated
Compliance Regulations Monitoring Tool	Eliminated
Okta SSO Authentication Module	Eliminated
Alteryx-Oracle Integration Module	Eliminated
Integration Monitoring Dashboard	Eliminated

Eliminated features specific workplace integration modules, anonymization of additional invoice training data, encryption tools, supplier notifications, and e-mail inbox connection.

2.3 User Characteristics

The intended user roles for AI^2 are Administrator, AP Team Member, AI Model Trainer, and Compliance Officer. Administrators configure AI^2 as needed to integrate it into the company's workspace, link it to the necessary single sign-on platforms and email inboxes, and be notified of system errors. AP Team Members use AI^2 to view their currently assigned invoices, verify PO matches, and make corrections as needed. AI Model Trainers monitor AI^2's performance in extracting invoice data fields and matching to the correct PO, prepare and add training data to improve the model's performance, and push model updates. Compliance officers will perform reviews of AI^2's audit logs and generate reports to ensure that the invoice handling adheres to internal standards and government regulations.

As AI^2 is intended for commercial use rather than personal use, users are expected to have professional-level expertise to complete their tasks. Usage scenarios include daily use by the AP team to verify invoice matches, use by AI model trainers to improve the model with additional training data, and periodic use by compliance officers to perform regulatory checks.

2.4 Constraints

Incorrect invoice and PO matching could compromise financial accuracy if it is not corrected, so confidence scoring will trigger human corrections before payment. The invoice parsing will not be 100% accurate for unusual formats and edge cases. The real-world product will require end-to-end encryption to ensure data confidentiality. Record retention and immutable audit logs will be required to comply with internal standards and federal law.

2.5 Assumptions and Dependencies

AI^2 requires an internet connection to use the web-based interface. It utilizes LiquidAI LFM 2.5 for invoice processing and matching, Flask-SQLAlchemy for database operations, and Tailwind CSS for frontend development.